

$$1. \quad f(a, b, c, d) = \sum m(2, 4, 10, 11, 12, 13)$$

$$f_1(a, b, c, d) = \prod M(0-3, 6-9, 12, 14, 15)$$

$$f_2(a, b, c, d) = \bar{b}c + a\bar{b}d$$

$$f_2 = \sum m(4, 5, 10, 11, 13) \leftarrow \text{SOP}$$

$$f_3 = \bar{b}c(a + \bar{a})(d + \bar{d}) + a\bar{b}d(c + \bar{c})$$

$$= \bar{b}c(ad + a\bar{d} + \bar{a}d + \bar{a}\bar{d}) + a\bar{b}cd + a\bar{b}\bar{c}d$$

$$= \bar{a}\bar{b}cd + a\bar{b}c\bar{d} + \bar{a}\bar{b}cd + \bar{a}\bar{b}c\bar{d} + a\bar{b}cd + a\bar{b}\bar{c}d$$

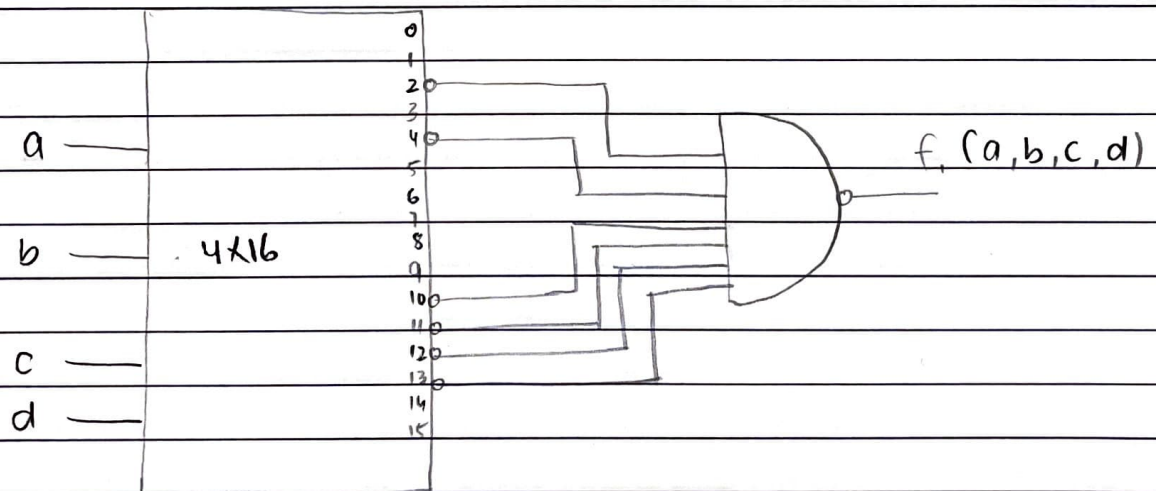
$$= 1011 \quad 1010 \quad 0011 \quad 0010 \quad 1011 \quad 1001$$

$$f_3 = \sum m(9, 2, 11, 10, 3)$$

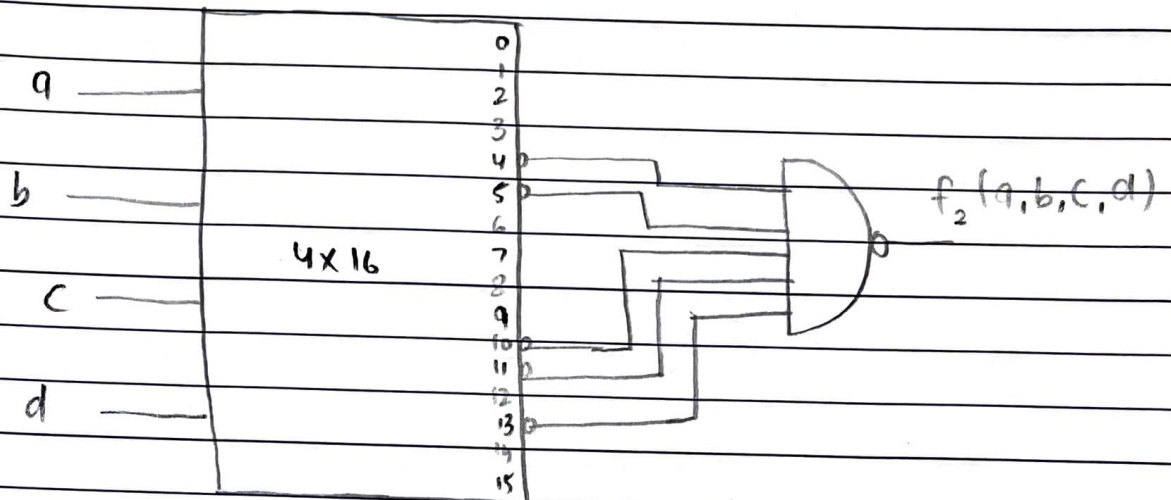
$$= \sum m(2, 3, 9, 10, 11)$$

Drawing circuits for each functions

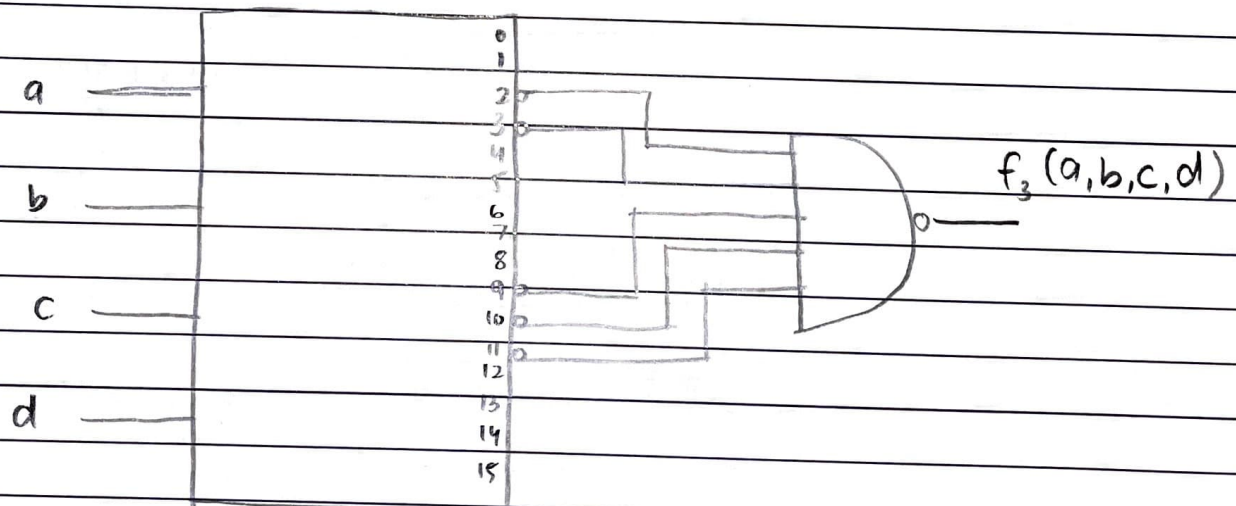
$$f(a, b, c, d) = \sum m(2, 4, 10, 11, 12, 13)$$



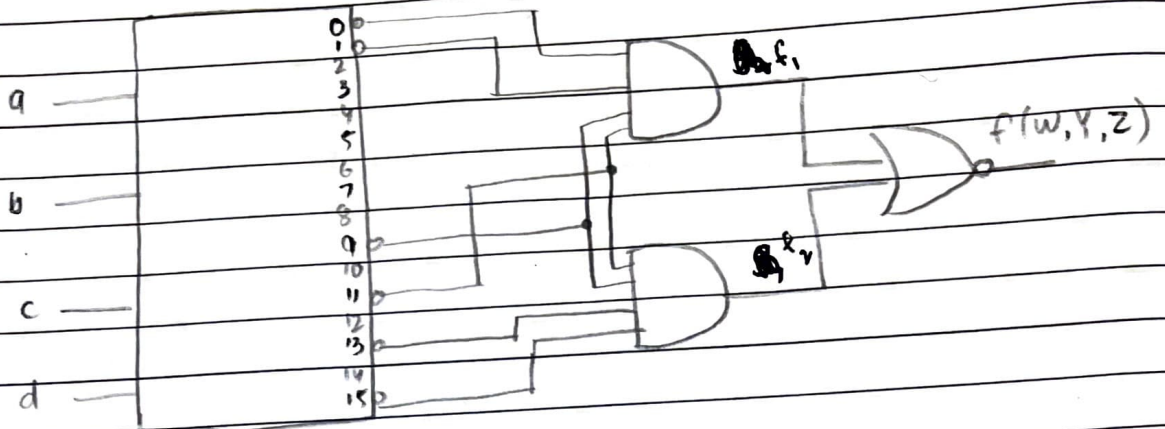
$$f_2(a, b, c, d) = \sum m(4, 5, 10, 11, 13)$$



$$f_3(a, b, c, d) = \sum m(2, 3, 9, 10, 11)$$



2.



$$f_1 = \Pi M(0, 1, 9, 11)$$

$$f_2 = \Pi M(9, 11, 13, 15)$$

wz \ yz	00	01	11	10
00	0	0		
01				
11				
10		0	0	

wx \ yz	00	01	11	10
00				
01				
11		0	0	
10		0	0	

$$f_1 = (w+x+y)(\bar{w}+x+\bar{z})$$

$$f_2 = (\bar{w} + \bar{z})$$

$$f = (f_1 + f_2)'$$

$$= (((w+x+y)(\bar{w}+x+\bar{z})) + \bar{w} + \bar{z})'$$

$$= ((w+x+y)(\bar{w}+x+\bar{z}))' \cdot \bar{w} \cdot \bar{z}$$

$$= ((w+x+y)' + (\bar{w}+x+\bar{z})') \cdot w \cdot z$$

$$= (\bar{w} \bar{x} \bar{y} + \bar{w} \bar{x} \bar{z}) w z$$

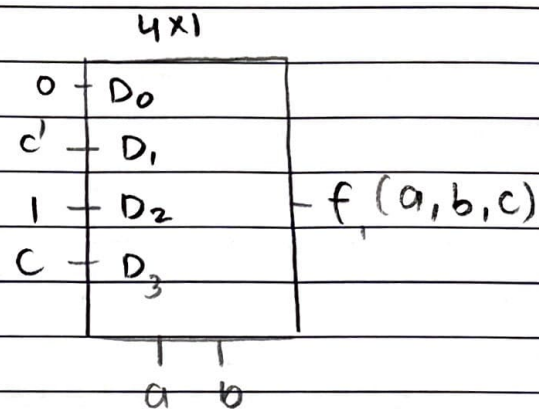
$$= \bar{w} \bar{x} \bar{y} w z + \bar{w} \bar{x} \bar{z} w z$$

$$= 0 + w \bar{x} z$$

$$= w \bar{x} z$$

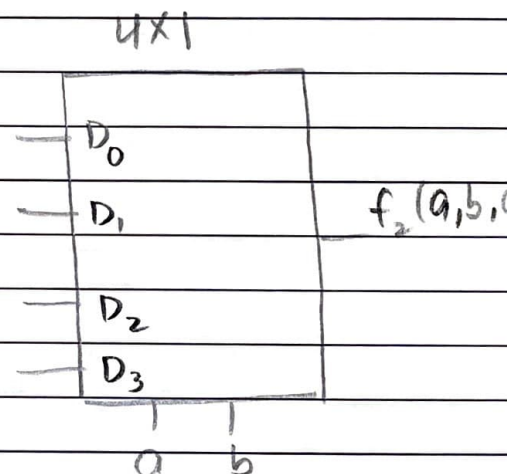
3. a. $f_1(a, b, c) = \sum m(2, 4, 5, 7)$

	A	B	C	Y
D_0	0	0	0	0 (0)
	0	0	1	0 (0)
D_1	0	1	0	1 (c')
	0	1	1	0 (c')
D_2	1	0	0	1 (1)
	1	0	1	1 (1)
D_3	1	1	0	0 (c)
	1	1	1	1 (c)



b. $f_2(a, b, c) = \prod M(0, 6, 7)$

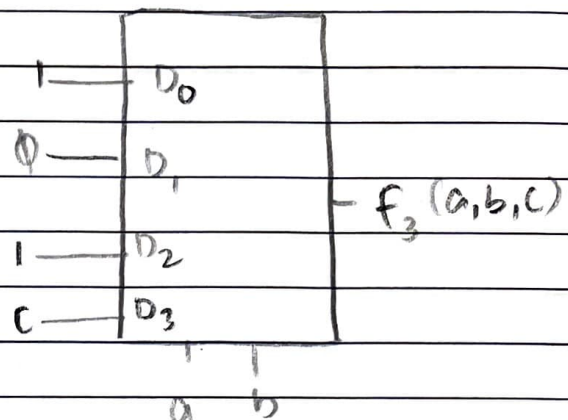
	A	B	C	Y
D_0	0	0	0	0 (c)
	0	0	1	1 (c)
D_1	0	1	0	1 (1)
	0	1	1	1 (1)
D_2	1	0	0	1 (1)
	1	0	1	1 (1)
D_3	1	1	0	0 (0)
	1	1	1	0 (0)



c. $f_3(a, b, c) = (a + \bar{b})(\bar{b} + c)$

$$\begin{aligned}
 &= a\bar{b} + ac + \bar{b}\bar{b} + \bar{b}c \\
 &= a\bar{b} + ac + \bar{b} + \bar{b}c \\
 &= ac + \bar{b} + \bar{b}c \\
 &= ac + \bar{b} \\
 &= ac(b + \bar{b}) + \bar{b}(a + \bar{a})(c + \bar{c}) \\
 &= abc + a\bar{b}c + a\bar{b}c + a\bar{b}\bar{c} + \bar{a}\bar{b}c + \bar{a}\bar{b}\bar{c} \\
 &= \sum m(0, 1, 4, 5, 7)
 \end{aligned}$$

	A	B	C	Y	
D_0	0	0	0	1	(1)
	0	0	1	1	(1)
D_1	0	1	0	0	(0)
	0	1	1	0	(0)
D_2	1	0	0	1	(1)
	1	0	1	1	(1)
D_3	1	1	0	0	(0)
	1	1	1	1	(1)



$$4. f(a,b,c) = \bar{a}\bar{b}\bar{c}D_0 + \bar{a}\bar{b}cD_1 + \bar{a}b\bar{c}D_2 + \bar{a}bcD_3 + a\bar{b}\bar{c}D_4 + a\bar{b}cD_5 + ab\bar{c}D_6 + abcD_7$$

and also,

$$m_0 = \bar{a}\bar{b}$$

$$m_1 = \bar{a}b$$

$$m_2 = a\bar{b}$$

$$m_3 = ab$$

From First
decoder

$$m_0 = \bar{c}\bar{b}$$

$$m_1 = \bar{c}b$$

$$m_2 = c\bar{b}$$

$$m_3 = cb$$

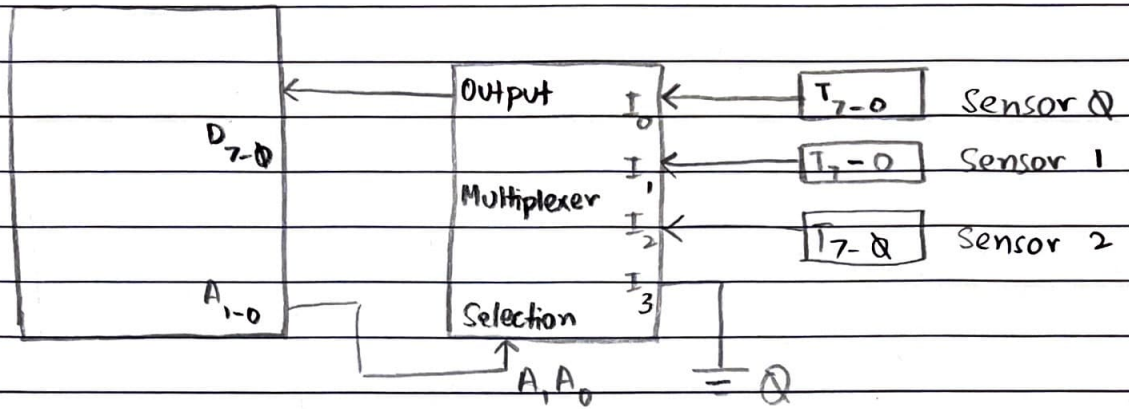
$$f(a,b,c) = \bar{a}\bar{b}\bar{c}(\bar{a}\bar{b}) + \bar{a}\bar{b}c(\bar{a}b) + \bar{a}b\bar{c}(a\bar{b}) + \bar{a}bc(ab) + a\bar{b}\bar{c}(\bar{c}\bar{b}) + a\bar{b}c(\bar{c}b) + ab\bar{c}(c\bar{b}) + abc(cb)$$

$$= \bar{a}\bar{b}\bar{c} + 0 + 0 + 0 + \bar{a}\bar{b}\bar{c} + 0 + 0 + abc$$

$$= \bar{a}\bar{b}\bar{c} + a\bar{b}\bar{c} + abc$$

$$f(a,b,c) = \sum m(0,4,7)$$

5.



In the multiplexer

